

Chapter 1 Elementary Theory of Univalent Mappings

1.1 Definitions

Definition 1.1.0.1

- A **single-valued** function f is said to be **univalent** in a domain

$$D \subseteq \mathbb{C}$$

if it never takes the same value twice. That is $f(z_1) \neq f(z_2)$ for all points z_1 and z_2 in D , with $z_1 \neq z_2$.

Remark It might be non-holomorphic!

- f is **locally univalent** if it is univalent in some neighborhood of z_0 ;
- Especially: An **analytic** univalent function is called **conformal mapping**, because of angle-preserving property.

We have

$$\text{Univalent} \Rightarrow \text{Locally Univalent}$$

but conversely may not hold, for example

♣ **Example 1.1.0.2** Consider the exponential map

$$t \mapsto e^{it}$$

It is locally univalent, because $f'(z_0) \neq 0$, $z_0 \in \mathbb{C}$. But it is not univalent because of period 2π .

More precisely:

♠ **Proposition 1.1.0.3** If f is analytic at z_0 , $f'(z_0) \neq 0$, then f is univalent in some neighborhood of z_0 ;
Conversely, if f is locally univalent, then $f'(z_0) \neq 0$.

◦ **Proof ...**

□

And according to

$$\text{Angle-Preserving} \Leftrightarrow f'(z) \neq 0$$

★ **Corollary 1.1.0.4** The univalent functions have angle-preserving property.

◦ **Proof ...**

□
